

Duc M. Nguyen

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Education

George Mason University, Ph.D. in Computer Science	Aug 2022 – May 2027
George Mason University, M.S. in Computer Science	Aug 2022 – Dec 2024
Miami University, B.S. in Computer Science	Aug 2018 – May 2022
• Minor in Statistics	GPA: 3.75/4.0

Experience

Research Assistant , RobotiXX Lab – Fairfax, VA	Aug 2022 – Now
• Building neural network-based contact prediction system to enable failure-aware motion planning for home service robots in dynamic environments. • Implemented motion planning algorithms and collision checking interface in MuJoCo, improving experimental efficiency and enabling rapid prototyping for research team. • Developed and deployed ROS-based autonomous navigation and manipulation system for mobile manipulators (Fetch, Spot) supporting multi-room task execution. • Engineered open-source wearable egocentric sensor suite integrating multiple modalities (RGB, depth, IMU) for large-scale human motion data capture. • Led collection and curation of MuSoHu dataset covering 100 km, 20 hours, 300 trials, and 13 participants across diverse public spaces, resulting in IROS 2023 publication.	
Research Engineering Intern , Johns Hopkins University – Baltimore, MD	May 2025 – August 2025
• Led team adoption of NVIDIA Isaac Sim/Lab, building GPU-accelerated RL training infrastructure supporting 10,000+ parallel environments with vision sensors for Unitree Go2 quadruped navigation. • Designed bilevel hierarchical control architecture separating high-level navigation from low-level locomotion, trained using deep reinforcement learning (PPO) in PyTorch and deployed on physical robot via ROS2. • Enabled Matterport3D indoor environments in Isaac Sim by developing automated mesh processing and lighting injection pipeline to convert incompatible simulation platform assets. • Implemented multi-sensor suite (LiDAR, IMU, RGB camera) matching physical robot specifications and designed observation spaces and reward structures for neural network policy training. • Set up AprilTag localization system throughout lab to enable vision-based odometry correction for mobile robots.	
Technical Lead , Google Developer Group at GMU – Fairfax, VA	Oct 2024 – August 2025
• Organized and moderated a university seminar featuring a Google engineer, conducting a live interview. • Planned and coordinated annual Google Developer Group events at George Mason University. • Led hands-on workshops and student projects introducing Google Cloud technologies and applied machine learning tools.	
Research Assistant , Miami University – Oxford, OH	May 2020 – Dec 2021
• Designed multi-agent deep reinforcement learning framework for cooperative UAV control under communication constraints, resulting in Springer book chapter publication. • Built machine learning models achieving 90% accuracy for behavioral data classification and developed interactive visualization tools using NetworkX and JavaScript.	

Selected Publications

Anticipatory Task and Motion Planning [\[PDF\]](#)

Roshan Dhakal, *Duc M. Nguyen*, Tom Silver, Xuesu Xiao, Gregory J. Stein

Human Uncertainty-Aware MPC for Enhanced Human-Robot Collaborative Manipulation [\[PDF\]](#)

Al Jaber Mahmud, *Duc M. Nguyen*, Filipe Veiga, Xuesu Xiao, Xuan Wang

IEEE 7th International Conference on Industrial Cyber-Physical Systems (ICPS), 2024

Toward Human-Like Social Robot Navigation: A Large-Scale, Multi-Modal, Social Human Navigation Dataset [\[PDF\]](#)

Duc M. Nguyen, Mohammad Nazeri, Amirrera Payandeh, Aniket Datar, Xuesu Xiao

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023

Responsive Regulation of Dynamic UAV Communication Networks Based on Deep Reinforcement Learning

R. Zhang, *Duc M. Nguyen*, M. Wang, L. X. Cai, X. Shen

Broadband Communications, Computing, and Control for Ubiquitous Intelligence. Wireless Networks. Springer, 2022

Projects

Energy-Based Diffusion Language Models for BabyLM

- Developing Energy-Based Diffusion Language Model (EDLM) combining discrete diffusion with sequence-level energy functions to improve sentence well-formedness on BabyLM benchmark dataset.
- Implementing compact diffusion model with contrastive learning objectives on **10M** word tracks, comparing performance against baseline autoregressive and standard diffusion models.

LinkedIn Job Posts Tracking

- Built an ETL pipeline that scrapes LinkedIn job posts into a MySQL database on an EC2 instance.
- Designed a S.T.A.R schema database to store over **1000** job postings using MySQL and Python.
- Extracted job details and requirements using regular expressions and Named Entity Recognition (NER).
- Delivered personalized job recommendations by aligning users' resumes with job postings.

Awards and Services

Finalist for Best Paper Award

AAAI Fall Symposium Artificial Intelligence for Human-Robot Interaction (AI-HRI), 2023

Journal/Conference Reviewer: ICRA, IEEE RA-L, AAAI AI-HRI, American Control Conference.

Technical Skills

Programming Languages: Python, C++ , Java, JavaScript, SQL

AI/ML Specializations: Reinforcement Learning, Diffusion Models, Energy-Based Models, Computer Vision, Optimization, Planning

AI/ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, LangChain, Google Cloud Vertex AI

Robotics Tools: ROS/ROS2, NVIDIA Isaac Sim/Lab, MuJoCo, Gazebo

Cloud & Databases: AWS (S3, EC2), Google Cloud Platform, MySQL, SQL Server

Development Tools: Git, Linux, macOS, Windows